Inoperability Assessment of Interdependent Critical Infrastructures by Minimal Inoperability Sets Analysis

Maria Valentina Clavijo Mesa
Energy Department
Politecnico di Milano
Milan, Italy
mariavalentina.clavijo@polimi.it

Francesco Di Maio*
Energy Department
Politecnico di Milano
Milan, Italy
francesco.dimaio@polimi.it
*Corresponding author

Enrico Zio
MINES Paris PSL University, Centre de Recherche sur les Risques et les Crises (CRC),
Sophia Antipolis, France
Energy Department, Politecnico di Milano
Milan, Italy
enrico.zio@polimi.it

***Abstract*—We consider the inoperability assessment of two Critical Infrastructures (CIs), one dependent on the other. Typically, such inoperability assessment is conducted by generating failures in the 'source' CI and modeling the disruption cascade into the dependent CI. In realistically large CIs, this results in the need of exploring a combinatorial number of accidental scenarios, which becomes infeasible due to the computational burden. In this paper, a novel approach is presented to address the computational issue, which allows identifying the Minimal Inoperability Sets (MISs), i.e., the minimal combinations of failures leading the dependent CI into a partial or fully inoperability state to exhaustively explore the accidental scenarios. A case study of a power CI feeding a water CI is considered.**

Keywords—Critical Infrastructures, Cascading Failures, Minimal Inoperability Sets, Dynamic Input-output Inoperability Model.

I. Introduction

Critical Infrastructures (CIs) like power grids, transportation networks, telecommunication systems and healthcare services are highly interconnected and their operability depend on each other's: when the operation of one CI is disrupted, it can affect the operability of another interconnected to it [1].

The Dynamic Input-Output Inoperability Model (DIIM) allows analyzing the dynamic behavior of interdependent CIs [2–4]. In [5], a computational approach is used to randomly sample failures in a CI and propagate their effects to the dependent CI. However, in real, large CIs this poses challenges for the exploration of the combinatorial number of possible disruption scenarios. This challenge can be overcome by the here proposed approach, which consists in first generating and simulating the accidental scenarios of failures in the dependent CI, and then identifying the Minimal Inoperability Sets (MISs), i.e., the minimal combinations of failures in the dependent CI that lead to a specific level of inoperability, and, finally using the DIIM to analyze the dynamics of inoperability.

A case study concerning a water network that depends on a power network is considered.

The remainder of the paper is organized as follows: Section 2 recall the DIIM for CI inoperability assessment, Section 3 describes the proposed approach, Section 4 introduces the case study, Section 5 presents the results and Section 6 concludes the work.

II. DIIM for inoperability assessment

DIIM describes the dynamic behavior of interdependent CIs in relation to their inoperability. The governing equation considering discrete time steps of one arbitrary unit of time, reads [6]:

$$\bar{q}(t+1) = \bar{q}(t) - \bar{\bar{K}}\bar{q}(t) + \bar{\bar{K}}\bar{\bar{A}}\bar{q}(t) + \bar{\bar{K}}\bar{c}(t) \tag{1}$$

where $\bar{q}(t)$ is the inoperability vector of the M interconnected CIs at time t, $\bar{\bar{K}}$ is a diagonal $M \times M$ matrix, where each entry k_{ii} represents the rate of recovery from disruption of the i-th CI, i.e., the speed at which each CI can recover from inoperability, $\bar{c}(t)$ is the perturbation vector at time t, whose entry $c_i(t)$ is the inoperability of the i-th CI due to direct effects of external perturbations, such as accidental events, natural disasters and targeted attacks, which cause disruptions, $\bar{\bar{A}}$ is a $M \times M$ matrix whose entry a_{ji} reflects the additional inoperability contributed by the i-th CI to the j-th CI (for example, if $a_{ji} = 1$, it means that complete inoperability of the i-th CI leads to complete inoperability of the j-th CI).

To estimate the interdependency coefficient a_{ji}, in [5] random disruptions are injected in the i-th CI and their impacts evaluated on the dependent j-th CI.

III. Proposed approach

The approach proposed to model cascading failures in interdependent CIs consists of the following five steps.

A. *Characterization of the CIs topology and operational attributes*

Each generic i-th CI ($i = 1, \ldots, M$) is represented as a direct graph $G^i \equiv \left(N^i, E^{(i,i)}\right)$, where $N^i = \left\{n_1^i, n_2^i, \ldots, n_Z^i\right\}$ are the Z

nodes (subsystems or components groups) of the i-th CI network, and edges $E^{(i,i)} = \left\{e_{op}^{(i,i)} = \left(n_o^i, n_p^i\right) \subseteq N^i \times N^j\right\}$ are the connections denoting either hardware or logical dependencies between the o-th and the p-th nodes of the i-th CI, G^i [7].

For each generic z-th node $n_z^i, (z = 1, \ldots, Z)$, a state variable vector $\bar{x}^{n_z^i} = \left[x_1^{n_z^i}, \ldots, x_{K_{n_z^i}}^{n_z^i}\right]$ defines its discrete multi-state $1, 2, \ldots, K_{n_z^i}$, where $x_1^{n_z^i}$ corresponds to fully functioning state of node n_z^i and $x_{K_{n_z^i}}^{n_z^i}$, to complete failure state [5].

When all nodes are in perfect state, $x_1^{n_z^i}\ \forall n_z^i \in N^i$, the generic i-th CI is in fully functional state X_1^i and the unmet demand of its service is 0. If the i-th CI is instead perturbed by a disruption, the states of its nodes suffer degradation and the unmet demand increases. The following performance metric, that uses demand satisfaction as a proxy for the overall operational state of the infrastructure, is adopted [8]:

$$O\left(X_l^i\right) = 1 - \frac{D(X_l^i)}{D(X_1^i)} \tag{2}$$

where $D(X_l^i)$ is the demand met by G^i in state X_l^i, and $D(X_1^i)$ is the attended demand of G^i in perfect functioning state X_1^i. The value range of the metric is $[0,1]$ for any state of the generic i-th CI: when $D\left(X_l^i\right) = D\left(X_1^i\right)$, G^i is fully functioning and $O\left(X_l^i\right) = 0$, denoting no inoperability; when $D\left(X_l^i\right) = 0$, G^i is unable to meet any demand and $O\left(X_l^i\right) = 1$ indicating complete inoperability.

B. Characterization of the dependencies

Dependencies can be physical, cyber, geographical, or logical relations between two CIs [9,10]. Here we consider that a generic j-th CI depends on the i-th CI (i.e., a dependency relation $i \to j$ exists), meaning that edges are unidirectionally connecting G^i and G^j, $E^{(i,j)} = \left\{e_{ol}^{(i,j)} = \left(n_o^i, n_l^j\right) \subseteq N^i \times N^j\right\}$. The sets of nodes involved in the dependency are denoted as: $N_{i\to j}^i$ and $N_{i\to j}^j$, where $N_{i\to j}^i \subseteq N^i$ contains those nodes of G^i directly connected to nodes in G^j, whereas $N_{i\to j}^j \subseteq N^j$ contains those nodes of G^j connected to G^i (see Figure 1).

We define the state sets $S_{i\to j}^i$ and $S_{i\to j}^j$ as the sets of states of the nodes in $N_{i\to j}^i$ and $N_{i\to j}^j$, respectively.

$$\begin{aligned} S_{i\to j}^i &= \left\{x_k^{n_z^i} \middle| n_z^i \in N_{i\to j}^i,\quad x_k^{n_z^i} \in \bar{x}^{n_z^i}\ for\ k = 1, \ldots, K_{n_z^i}\right\} \\ S_{i\to j}^j &= \left\{x_k^{n_z^j} \middle| n_z^j \in N_{i\to j}^j,\quad x_k^{n_z^j} \in \bar{x}^{n_z^j}\ for\ k = 1, \ldots, K_{n_z^j}\right\} \end{aligned} \tag{3}$$

The so called dependency matrix $\bar{\bar{M}}_{i\to j}$ is, then, populated: if a given $S_{i\to j}^i$ (i.e., the given states of the nodes in $N_{i\to j}^i$) leads to a given $S_{i\to j}^j$ (i.e., given states of the nodes in $N_{i\to j}^j$), then $\bar{\bar{M}}_{i\to j}\left[S_{i\to j}^i, S_{i\to j}^j\right] = 1$, and otherwise 0.

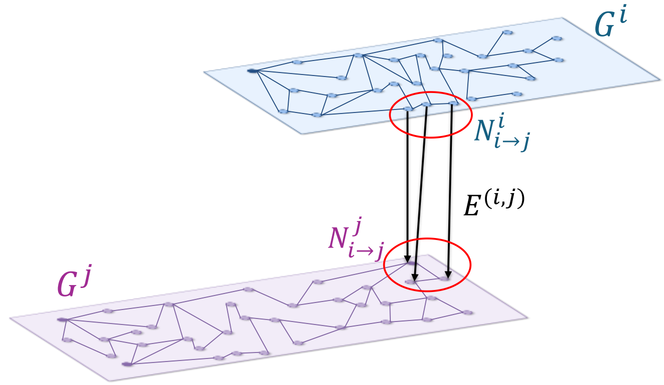


Figure 1. Example of dependency of G^j on G^i

C. Exploration of scenarios in the dependent CI

To explore the disruption scenarios in G^i that cascade into G^j, an exhaustive set of scenarios $\Omega^j = \left\{\xi_1^j, \xi_2^j, \ldots, \xi_F^j\right\}$ is considered, each scenario being defined by sets of $S_{i\to j}^j$ degraded/failed for which $\bar{\bar{M}}_{i\to j}$ entries are 1 (i.e., there is dependency).

Figure 2 exemplifies the Ω^j for Figure 1, assuming that, for simplicity but without loss of generality, degraded states of nodes in $N_{i\to j}^i$ result in complete failure of nodes in $N_{i\to j}^j$.

For each scenario ξ_f^j, the attended demand $D\left(X_{\xi_f^j}^j\right)$ and, consequently, the performance of the infrastructure under that scenario $O\left(X_{\xi_f^j}^j\right)$ are calculated and recorded in a performance vector $\bar{O}_{i\to j}^j$.

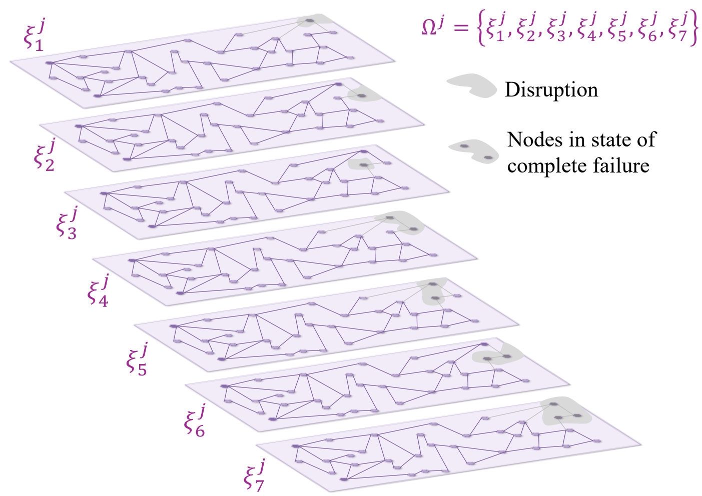


Figure 2. Example of Ω^j failure scenarios generation for G^j

D. Minimal Inoperability Sets (MISs)identification

An iterative algorithm (see pseudocode in Table I) searches for the minimal combination of node failures that lead to each inoperability interval r_l^j, where $r_l^j = \{[c_l, d_l]\}\ \ (l = 1, \ldots, r^j$,

$0 = c_1 < d_1 = c_2 < d_2 \ldots = c_{r^j} < d_{r^j} = 1$), is a discretized inoperability interval of the j-th CI. In practice, the algorithm filters out non-minimal sets of the minimal ones to mine out the MISs: ξ_f^j is considered a non-minimal of ξ_h^j if the elements of ξ_h^j are contained in ξ_f^j.

TABLE I. PSEUDOCODE TO IDENTIFY MISS

Input to define MISs given the dependency $i \rightarrow j$:
Inoperability intervals of the dependent j-th CI, r^j
Performance vector, $\bar{O}_{i\rightarrow j}^j$
Set of failure scenarios, Ω^j

for each inoperability interval $r_l^j \in r^j$:
    search ξ_f^j in Ω^j that result in a performance $O_{\xi_f^j}^j \in r_l^j$
    define f_l^j as the failure scenarios that results in r_l^j
end
for each failure scenarios vector f_l^j:
    initialize $\bar{M}_l^j = \{\emptyset\}$ as minimal inoperability sets of intervals r^j
    for each failure scenario $\xi_f^j \in f_l^j$:
        set is_non_minimal = False
        if $\xi_f^j \supset \xi_h^j \in \bar{M}_l^j$:
            return is_non_minimal = True
            break
        else:
            store ξ_f^j in $\bar{M}_l^j$
        end
    end
end

Output: $\bar{M}_l^j$, for each inoperability interval $r_l^j \in r^j$ of the dependent j-th CI

E. Estimation of inoperability by DIIM

The inoperability of each CI is estimated using Equation (1), whose $a_{ji}(r_e^i)\ \forall r_e^i \in r^i$ are calculated as:

$$a_{ji}(r_e^i) = \sum_{l=1}^{r^j} P\left(O\left(X_u^j\right) \in r_l^j \middle| O\left(X_p^i\right) \in r_e^i\right) \times d_l \tag{4}$$

where d_l is, as already mentioned, the upper limit of r_l^j, so that we conservatively consider the worst impact of the disruption on the j-th CI, and $P\left(O\left(X_u^j\right) \in r_l^j \middle| O\left(X_p^i\right) \in r_e^i\right)$ is the conditional probability, calculated using the identified MISs, that the operational performance of the j-th CI in state X_u^j falls within the interval r_l^j, given that the operational performance of the i-th CI, in state X_p^i is within the interval $r_e^i, r_e^i = \{[a_e, b_e]\}$ ($e = 1, \ldots, r^i, 0 = a_1 < b_1 = a_2 < b_2 \ldots = a_{r^i} < b_{r^i} = 1$).

In particular, for a generic s-th disruption scenario ξ_s^i of the i-th CI, if it does not lead to the occurrence of any MIS on the j-th CI, the operational performance of G^j is automatically set $O\left(X_1^j\right) = 0$; otherwise, the operational performance of G^j is assigned to the corresponding inoperability interval r_l^j.

IV. CASE STUDY

We consider a water CI dependent on a power CI. The power network (PN) is based on the IEEE57-bus test system and the topology of the water supply system (WN) is taken from the IEEE85-bus system [11]. We distinguish the nodes of these CIs as supplier and demand nodes, based on their role in the flow of commodities [12,13]. Then, the relevant nodes sets are $N^{PN} = N_S^{PN} \cup N_D^{PN}$ and $N^{WN} = N_S^{WN} \cup N_D^{WN}$.

Supplier nodes (sets N_S^{PN} and N_S^{WN}) can be in four states: perfect functioning $\left(x_1^{n_z^{PN}}, x_1^{n_z^{WN}} \forall n_z^{PN} \in N_S^{PN}, n_z^{WN} \in N_S^{WN}\right)$, disconnected (due to loss of edges connecting to demand nodes) $\left(x_2^{n_z^{PN}}, x_2^{n_z^{WN}}\right)$, overloaded (exceeding capacity) $\left(x_3^{n_z^{PN}}, x_3^{n_z^{WN}}\right)$ or inoperable $\left(x_4^{n_z^{PN}}, x_4^{n_z^{WN}}\right)$. Demand nodes (sets N_D^{PN} and N_D^{WN}) can be in three states: perfect functioning $\left(x_1^{n_z^{PN}}, x_1^{n_z^{WN}} \forall n_z^{PN} \in N_D^{PN}, n_z^{WN} \in N_D^{WN}\right)$, disconnected (due to loss of edges connecting to supplier nodes) $\left(x_2^{n_z^{PN}}, x_2^{n_z^{WN}}\right)$ or inoperable $\left(x_3^{n_z^{PN}}, x_3^{n_z^{WN}}\right)$.

The PN topology, G^{PN}, includes a set N_S^{PN} of seven supply nodes providing electricity to fifty demand nodes in set N_D^{PN}. In its perfect functioning state, X_1^{PN}, the provides power to meet a demand of $D(X_1^{PN}) = 428.8$ MW. The WN topology, G^{WN}, in its perfect functioning state X_1^{WN} delivers water flow for a total demand $D(X_1^{WN}) = 2514.28\ m^3/h$ from only one node in set N_S^{WN} to the eighty-four nodes of set N_D^{WN} (see Figure 3).

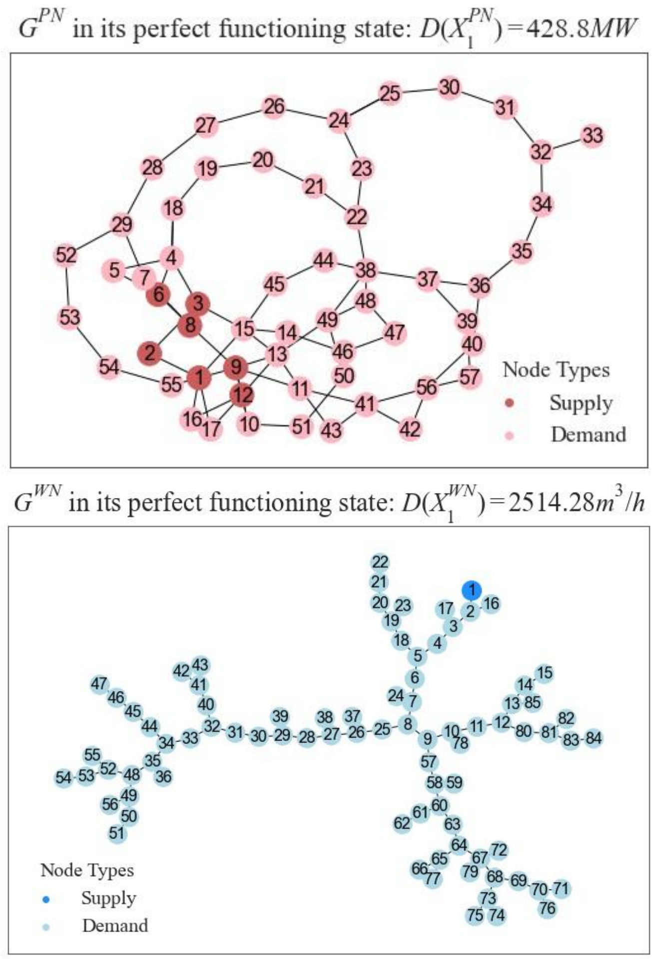


Figure 3. Topologies of G^{PN} and G^{WN} for the case study

The dependency $PN \rightarrow WN$ implies that water distribution through WN requires power supplied by PN. This physical dependency is provided by the 15 nodes in G^{PN} supplying 11 demand nodes in G^{WN} (see Figure 4).

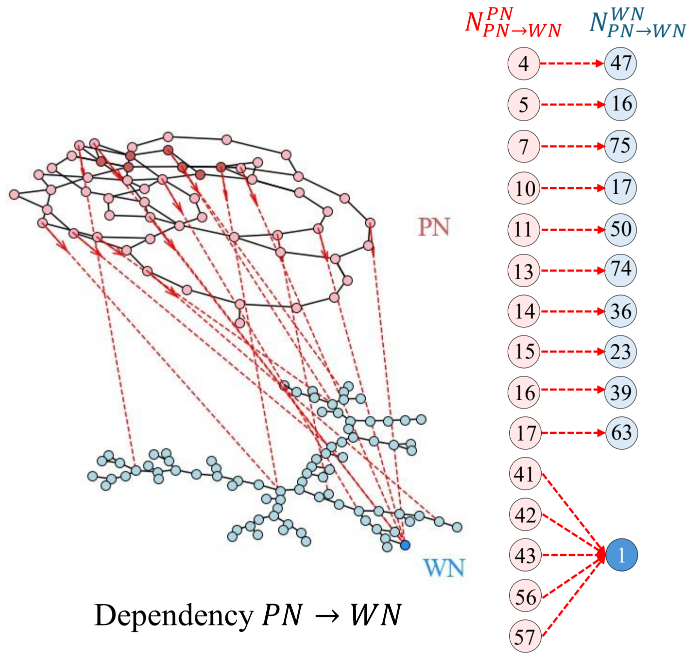


Figure 4. Physical dependency $PN \rightarrow WN$

The dependency matrix $\bar{\bar{M}}_{PN\rightarrow WN}$ is constructed using the state sets $S^{PN}_{PN\rightarrow WN}$ and $S^{WN}_{PN\rightarrow WN}$. The matrix is filled as follows: if a node in $N^{PN}_{PN\rightarrow WN}$ results in a state different of perfect functioning, then the corresponding node in $N^{WN}_{PN\rightarrow WN}$ is inoperable and $\bar{\bar{M}}_{PN\rightarrow WN}[S^{PN}_{PN\rightarrow WN}, S^{WN}_{PN\rightarrow WN}] = 1$. The supplier node in $N^{WN}_{PN\rightarrow WN}$, supported by five nodes in $N^{PN}_{PN\rightarrow WN}$, become inoperable only if all five supporting nodes are in a state different to perfect functioning.

V. RESULTS

The set Ω^{WN} consists of 2047 disruption scenarios. The inoperability of WN is discretized in five intervals:

$$r^{WN} = [0], (0, 0.2], (0.2, 0.6], (0.6, 0.8], (0.8, 1] \quad (5)$$

The MISs leading to the different intervals of inoperability are listed in Table II. This allows considering only the 18 MISs in WN instead of evaluating by DIIM the cascading failure process for all 2047 disruption scenarios.

The inoperability range of PN is also discretized as follows:

$$r^{PN} = \{[0], (0, 0.3], (0.3, 0.5], (0.5, 0.7], (0.7, 1]\} \quad (6)$$

The estimated interdependency coefficients for each inoperability interval of PN are shown in Table III. Weighted by the frequencies plotted in the histogram, they provide the $a_{WNPN}[r^{PN}_k]$ to be used in the interdependency matrix, as in Equation (7), and to be updated based on the inoperability of PN at time t:

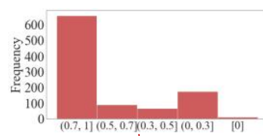
$$\bar{A}(t+1) = \begin{pmatrix} 0 & a_{WNPN}(q_{PN}(t)) \\ 0 & 0 \end{pmatrix} \quad (7)$$

Assuming arbitrary recovery periods of 14 days for PN and 21 days for WN, we use Equation (1) to estimate the inoperability of the two CIs.

TABLE II. MINIMAL INOPERABILITY SETS OF WN

r^{WN}	MISs
$r_1^{WN} = [0]$	$\bar{M}_1^{WN} = \{\emptyset\}$
$r_2^{WN} = (0, 0.2]$	$\bar{M}_2^{WN} = \left\{ \left[x_3^{n_{16}^{WN}}\right], \left[x_3^{n_{17}^{WN}}\right], \left[x_3^{n_{23}^{WN}}\right], \left[x_3^{n_{36}^{WN}}\right], \left[x_3^{n_{39}^{WN}}\right], \left[x_3^{n_{47}^{WN}}\right], \left[x_3^{n_{50}^{WN}}\right], \left[x_3^{n_{63}^{WN}}\right], \left[x_3^{n_{74}^{WN}}\right], \left[x_3^{n_{75}^{WN}}\right] \right\}$
$r_3^{WN} = (0.2, 0.6]$	$\bar{M}_3^{WN} = \left\{ \left[x_3^{n_{16}^{WN}}, x_3^{n_{63}^{WN}}\right], \left[x_3^{n_{17}^{WN}}, x_3^{n_{63}^{WN}}\right], \left[x_3^{n_{23}^{WN}}, x_3^{n_{63}^{WN}}\right], \left[x_3^{n_{36}^{WN}}, x_3^{n_{63}^{WN}}\right], \left[x_3^{n_{39}^{WN}}, x_3^{n_{63}^{WN}}\right], \left[x_3^{n_{47}^{WN}}, x_3^{n_{63}^{WN}}\right], \left[x_3^{n_{50}^{WN}}, x_3^{n_{63}^{WN}}\right] \right\}$
$r_4^{WN} = (0.6, 0.8]$	$\bar{M}_4^{WN} = \{\emptyset\}$
$r_5^{WN} = (0.8, 1]$	$\bar{M}_5^{WN} = \left\{ \left[x_3^{n_1^{WN}}\right] \right\}$

TABLE III. INTERDEPENDENCY COEFFICIENTS FOR G^{PN}



r^{PN} \ r^{WN}	[0]	(0, 0.2]	(0.2, 0.6]	(0.6, 0.8]	(0.8, 1]	$a_{WNPN}[r^{PN}_k]$
[0]	$1 \times 0 +$	$0 \times 0.2 +$	$0 \times 0.6 +$	$0 \times 0.8 +$	$0 \times 1 =$	0
(0, 0.3]	$0.41 \times 0 +$	$0.50 \times 0.2 +$	$0.08 \times 0.6 +$	$0 \times 0.8 +$	$0.01 \times 1 =$	0.16
(0.3, 0.5]	$0.03 \times 0 +$	$0.69 \times 0.2 +$	$0.20 \times 0.6 +$	$0 \times 0.8 +$	$0.08 \times 1 =$	0.34
(0.5, 0.7]	$0.01 \times 0 +$	$0.47 \times 0.2 +$	$0.35 \times 0.6 +$	$0 \times 0.8 +$	$0.17 \times 1 =$	0.47
(0.7, 1]	$0 \times 0 +$	$0.04 \times 0.2 +$	$0.06 \times 0.6 +$	$0 \times 0.8 +$	$0.90 \times 1 =$	0.94

Figure 5 shows the inoperability of the CIs under two disruption scenarios: a high disruption with PN at 70% initial inoperability, and a medium disruption at 30%. In the high disruption scenario, the interdependency coefficient $a_{WNPN}(q_{PN}(t))$ at time $t = 0$ is 0.47 reflecting a significant impact on WN. In the medium disruption scenario, it reduces to $a_{WNPN}(q_{PN}(0)) = 0.16$ due to the lower initial inoperability interval of PN (r_2^{PN}). These scenarios illustrate the influence of the dependency $PN \rightarrow WN$ on the cascading process into WN.

Table IV compares the values obtained for each inoperability interval of r^{PN} using the approach detailed in [5] and the one here proposed. For each approach, 1000 disruption scenarios are considered: in the former approach, 1000 random disruption scenarios were sampled in the PN, followed by the propagation of cascading failures in the WN, whereas in the proposed approach, 1000 randomly sampled disruption scenarios are compared with MISs to establish the state of WN, without the need to simulate the cascading failure process.

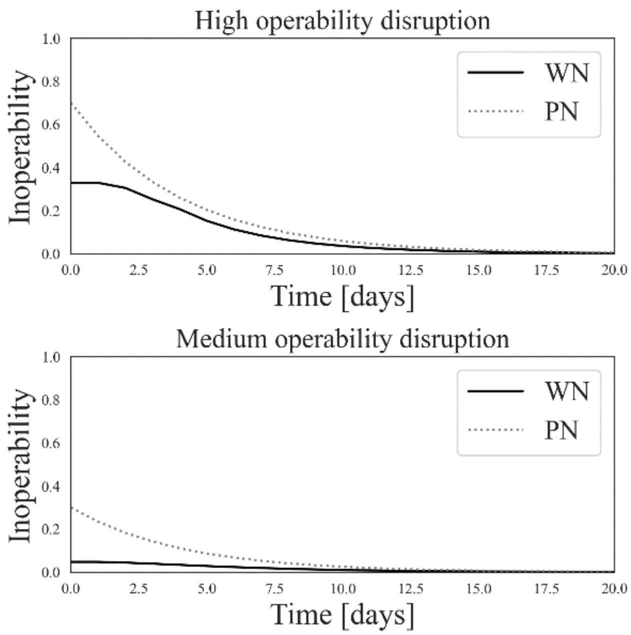


Figure 5. DIIM for two disruption scenarios

TABLE IV. INTERDEPENDENCY COEFFICIENTS ESTIMATED USING [5] AND PROPOSED APPROACH

		without MISs [5]	with MISs (proposed)
		$a_{WNPN}[r_k^{PN}]$	$a_{WNPN}[r_k^{PN}]$
r^{PN}	$r_1^{PN} = [0]$	0	0
	$r_2^{PN} = (0, 0.3]$	0.18	0.16
	$r_3^{PN} = (0.3, 0.5]$	0.37	0.34
	$r_4^{PN} = (0.5, 0.7]$	0.60	0.47
	$r_5^{PN} = (0.7, 1]$	0.94	0.94

Both approaches yield similar numerical results. However, as shown in Table V, the proposed approach is faster in analyzing the cascading failure propagation in the CIs. In fact, the proposed approach requires 66% of the computational time spent by the approach in [5], using 13.02 seconds to identify the MISs of the WN and 48.08 seconds to estimate the interdependency coefficients. Additionally, the proposed approach identifies MISs in the dependent CI, offering decision-makers relevant information about the minimal combinations of node states that ensure specific performance levels in the infrastructure. This information can be valuable for planning mitigation and contingency strategies after disruptions.

VI. CONCLUSIONS

This paper presents a novel approach for assessing the inoperability of multi-state dependent CIs. The approach performs the analysis of accidental scenarios in dependent infrastructures by focusing on the identification of MISs, availing to fully explore cascading failure effects, thus providing reduction of the computational burden. A case study involving a power network and a water network that depends on power supply for its operation exemplifies the proposed approach and its benefits.

TABLE V. COMPARISON BETWEEN [5] AND PROPOSED APPROACHES

Characteristic	[5]	Proposed
Identification of the most frequent inoperability interval of each CI	✓	✓
Ability to model the stochastic nature of dependency relation between CIs	✓	✓
Identification of minimal node combinations (MISs) that ensure each inoperability interval in the dependent CI	✗	✓
Computational time for 1000 disruption scenarios	92.82 seconds	61.10 seconds

*Both approaches were analyzed on an Intel Core i5-11400H CPU, 16 GB RAM.

ACKNOWLEDGMENT

The work of Maria Valentina Clavijo Mesa has been financed by - Next Generation EU, Missione 4, componente 1 “Potenziamento dell’offerta dei servizi di istruzione: dagli asili nido all’Università” – Investimento 3.4 “Didattica e competenze universitarie avanzate” e Investimento 4.1 “Estensione del numero di dottorati di ricerca e dottorati innovativi per la pubblica amministrazione e il patrimonio culturale” CUP D43C22002930001.

REFERENCES

[1] Wang Y, Guo P, Wu Y, Chen Y, Zio E. Vulnerability assessment of interdependent infrastructures based on a cascading failure model. 2022 6th International Conference on System Reliability and Safety (ICSRS), IEEE; 2022, p. 40–4. https://doi.org/10.1109/ICSRS56243.2022.10067571.

[2] Jin J, Zhou H. A Demand-side inoperability Input–Output Model for strategic risk management: insight from the COVID-19 outbreak in Shanghai, China. Sustainability 2023;15:4003. https://doi.org/10.3390/su15054003.

[3] Setola R, De Porcellinis S, Sforna M. Critical infrastructure dependency assessment using the input-output inoperability model. International Journal of Critical Infrastructure Protection 2009;2:170–8. https://doi.org/10.1016/j.ijcip.2009.09.002.

[4] Santos J, Roquel KIDZ, Lamberte A, Tan RR, Aviso KB, Tapia JFD, et al. Assessing the economic ripple effects of critical infrastructure failures using the dynamic inoperability input-output model: a case study of the Taal Volcano eruption. Sustain Resilient Infrastruct 2023;8:68–84. https://doi.org/10.1080/23789689.2022.2127999.

[5] Clavijo-Mesa MV, Di Maio F, Zio E. ‘unpublished’ Dynamic Inoperability Input-Output modeling of a system of systems made of multi-state interdependent critical infrastructures. Under revision. Reliab Eng Syst Saf 2024.

[6] Haimes YY, Jiang P. Leontief-based model of risk in complex interconnected infrastructures. Journal of Infrastructure Systems 2001;7:1–12.

[7] Bompard E, Pons E, Di Wu. Extended topological metrics for the analysis of power grid vulnerability. IEEE Syst J 2012;6:481–7. https://doi.org/10.1109/JSYST.2012.2190688.

[8] Wang K, Xu Z, Liu Y, Fang Y. Resilience enhancement for multistate interdependent infrastructure networks: from a preparedness perspective. IEEE Trans Reliab 2023;72:190–203. https://doi.org/10.1109/TR.2021.3132774.

[9] Setola R, Rosato V, Kyriakides E, Rome E. Managing the complexity of critical infrastructures: A modelling and simulation approach. 2016.

[10] Wolfgang Kroger, Enrico Zio. Chapter 2: Properties of critical infrastructures. Vulnerable Systems, 2011, p. 9–30.

[11] MATPOWER. Case 57 Power flow data for IEEE 57 bus test case. Https://MatpowerOrg/Docs/Ref/Matpower50/Case57Html 2014.

[12] Popov I. Multi-supplier power grid framework based on multicommodity routing. 2015 International Energy and Sustainability Conference (IESC), IEEE; 2015, p. 1–7. https://doi.org/10.1109/IESC.2015.7384387.

[13] Rubido N, Grebogi C, Baptista MS. Resiliently evolving supply-demand networks. Phys Rev E 2014;89:012801. https://doi.org/10.1103/PhysRevE.89.012801.